Project 2: Wind Turbine Design

MECH 3100: Fluid Mechanics

Mia Vazquez, Kasin Ito-Sato, Andy Salgado

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Design overview

The airfoil that was used for the wind turbine design was NACA0012-sa with a 20mm chord and angle of attack of 12.5 degrees. This NACA airfoil was chosen because it is a simple, symmetric 4-digit airfoil with well documented lift characteristics, making it straightforward to model, analyze, and manufacture for a small scale wind turbine. The final design used 4 airfoils and with the chosen angle of attack of 12.5 degrees the lift coefficient is approximately 1.1064 [1]. In the images below are the calculations for lift force, power from the wind turbine, and power generated from the motor. The wind turbine efficiency and calculated Reynolds number are also included, along with some early stage design sketches of the wind turbine design.

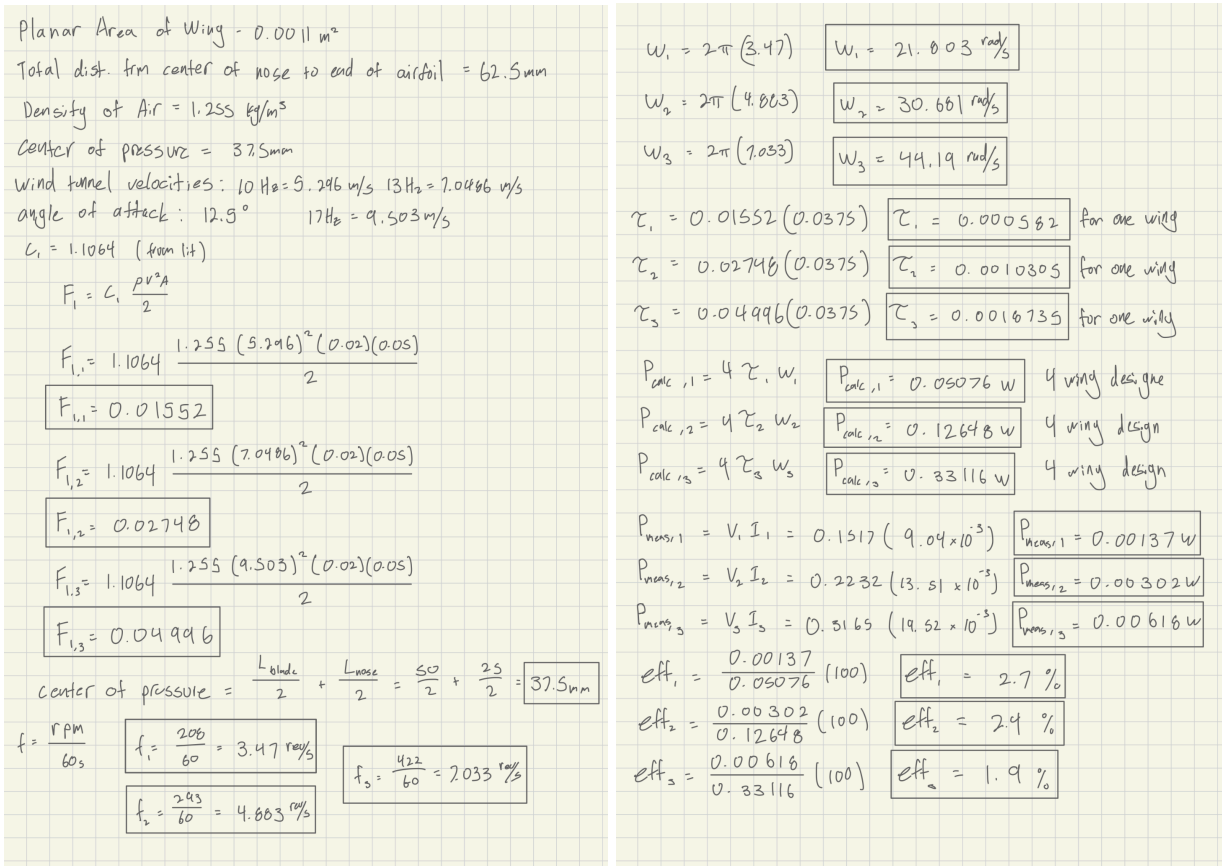


Figure 1 & 2: Lift Force, Angular velocity, Torque, Mechanical Power, Electrical Power, and Efficiency Calculations.

$$Re = \frac{\rho V c}{\mu}$$

$$Re_1 = \frac{1.255\,(5.296)\,(0.02)}{1.789 \times 10^{-5}} \qquad \boxed{Re_1 = 7430.39}$$

$$Re_2 = \frac{1.255\,(7.0486)\,(0.02)}{1.789 \times 10^{-5}} \qquad \boxed{Re_2 = 9889.32}$$

$$Re_3 = \frac{1.255\,(9.503)\,(0.02)}{1.789 \times 10^{-5}} \qquad \boxed{Re_3 = 13332.88}$$

Figure 3: Reynolds Number Calculation

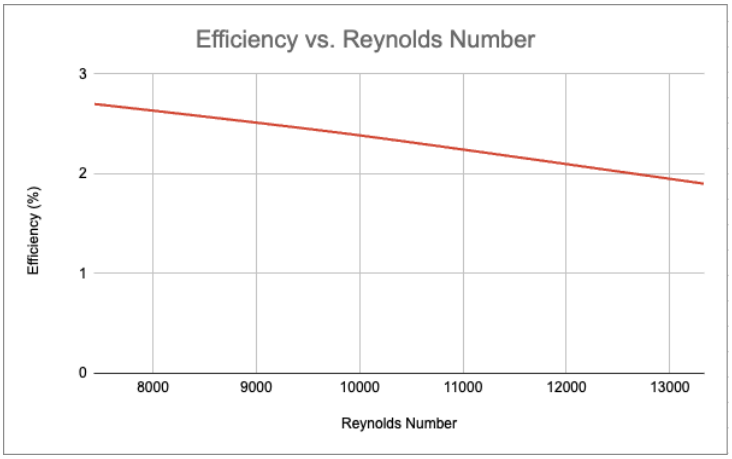
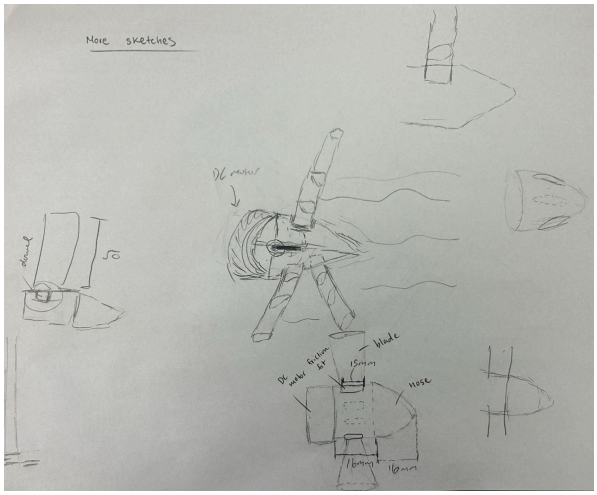
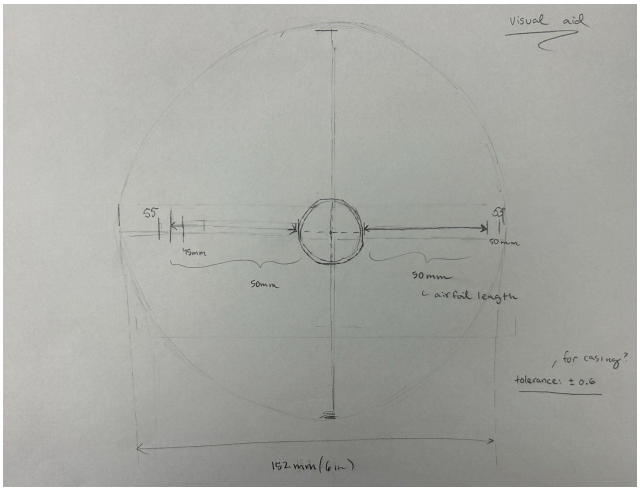


Figure 4: Reynolds Number vs. Wind Turbine Efficiency Plot





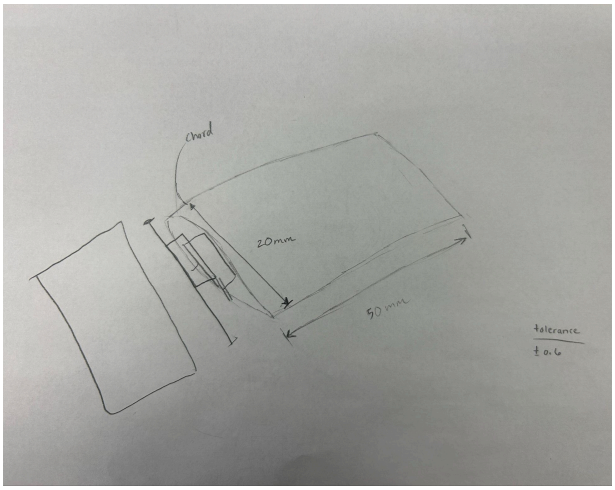


Figure 5, 6, & 7: Air Foil Sketches

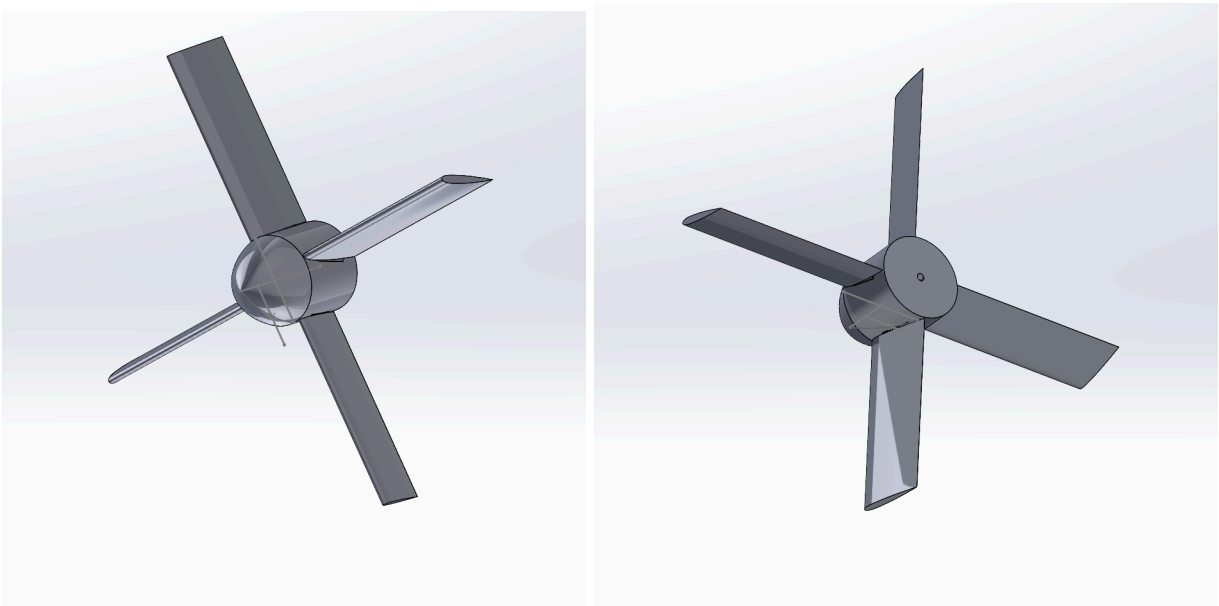


Figure 8 & 9: Wind Turbine airfoils and Nose SolidWorks Assembly



Figure 10: Completed Wind Turbine

Results

The calculated power values for each trial, shown as P_{calc}, were found using the expression $P = \tau\omega$, where torque was calculated by multiplying the center of pressure by the lift force of each trial, and where the frequency of each trial was multiplied by 2π to find the rotational speed. The calculated results were:

$$P_{calc,1} \approx 0.05076\ W$$

$$P_{calc,2} \approx 0.12648\ W$$

$$P_{calc,3} \approx 0.33116\ W$$

The measured power values were then found by multiplying the measured current by the voltage acquired during the experiment:

$P_{meas,1} \approx 0.00137\,W$

$P_{meas,2} \approx 0.00302\,W$

$P_{meas,3} \approx 0.00618\,W$

The measured power produced by the DC motor is much lower than the calculated power, showing that the actual airfoil system delivers much less power than the theoretical model. The calculated model does not account for friction in the DC motor or drag that formed, and is a much more ideal result. The actual performance of our airfoil system was reflected by the measured values. When comparing, we can see that the real system does not operate near the ideal system, and it is apparent the calculated power does not show the effects of inefficiencies or friction.

Using the SolidWorks simulation, we were able to display the velocity distribution around the airfoil at a 12.5° angle of attack, and the velocity used was 9.503 m/s. The contour plot shown in Figure 11 shows that the airflow speeds up significantly over the upper surface, shown by the red and orange regions, while the lower surface shows lower velocities. There is a low-velocity wake that forms, shown by the regions in blue, which indicates increased drag.

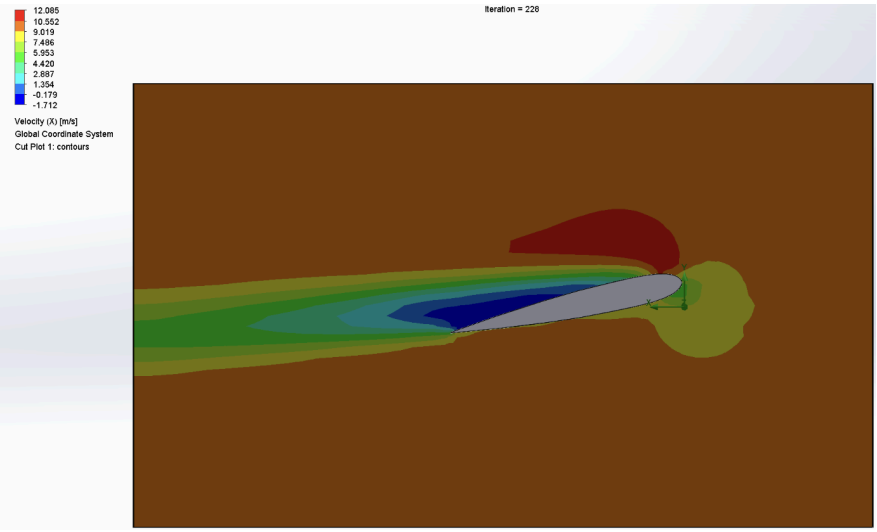


Figure 11: Cutplot of Velocity Distribution Around Airfoil

The pressure contour plot (Figure 12) shows a pressure distribution at the 12.5° angle of attack. There is a lower pressure region shown by the blue/teal patch on the upper surface of the airfoil, indicating a static pressure drop which generates lift due to the higher pressure around the lower surface of the airfoil. At the very front edge of the airfoil, there is a small area of elevated pressure created by the stagnation point that forms behind the airfoil, aligning with the velocity results in Figure 11. These trends demonstrate how the airfoil geometry affects how lift and wake effects are generated.



Figure 12: Cut Plot of Pressure Distribution Around Airfoil

Together, the velocity and pressure contour plots show how the airfoil generates lift at a positive angle of attack. The velocity contour indicates that airflow accelerates over the upper surface, corresponding to the pressure drop region in the static pressure plot. When the lower surface faces a lower velocity with a higher pressure, a lift is created due to the change in pressure. A wake is shown in both plots, with low velocity shown and caused by the pressure difference of higher and lower pressures on the lower and upper parts of the airfoil respectively. There is a loss of momentum which shows up on the simulation as a drop in velocity which indicates increased drag.

Conclusions

Overall, there were clear differences between theoretical and experimental values for our turbine's performance. The calculated power was consistently higher than the measured power due to possible losses such as a DC motor inefficiency, friction, drag, and flow conditions shown in the SolidWorks simulation. It is also possible the 3D-printed blades caused an imbalance, either due to different fill settings or filing the blades themselves, which reduced stability during

our wind tunnel trials. The airfoil angle also strongly affected the lift. It is possible that we were able to increase the angle of attack to increase the surface area and lift, but too much angle might have introduced more drag to the system, reducing efficiency. To improve our design in the future, we could experiment using more precise manufacturing methods to reduce imbalance in the system, such as laser cutting acrylic or wood. It would be important to also consider lighter and stiffer materials to improve the aerodynamics of the system. These changes would increase the turbine stability, help to improve efficiency, and bring our measured power closer to the calculated power predictions.

Appendix A

Table A1: Collected Wind Tunnel and Voltmeter Data

RPM	Volts	miliAMPs	Hertz
208	0.1517	9.04	10
293	0.2232	13.51	13
422	0.3165	19.52	17

Table A2: Measured Wind Tunnel Velocities

Hz	m/s
10	5.296
13	7.0486
17	9.503

References

[1] http://airfoiltools.com/polar/details?polar=xf-naca0012h-sa-100000-n5.